EV Charger Permit Packet

123 Sample St, Oro Valley, AZ

JOB	CONTRACTOR	CHECKLIST
2,100 sq ft home Parcel 000-00-000-000 Charger at attached garage Customer: Sample Homeowner	Sample Electric, LLC License ROC 000000	Oro Valley, AZ From the city's published checklist

Checklist

Required	In this packet	Status
Completed building permit application, with Oro Valley business license and AZ registrar numbers	Page 2: fields ready for the city's form	**Ready**
Site plan showing the charger and main electrical panel	Page 3: to scale, 1 in = 30 ft	**Ready**
Charger design and UL listing information	Page 6: add manufacturer sheets	**Attach**
One-line diagram with conductors, conduit sizes and types, and overcurrent protection	Page 4	**Ready**
Load calculation for a new 240 V circuit	Page 5	**Ready**
Venting shown on the plans, if the charger requires it	Only if this charger requires venting	**Check**
HOA or subdivision rules	Homeowner to confirm before work starts	**Open**

Engineering checks

Check	Result	Status
Service capacity	165.8 A computed on a 200 A service	**Pass**
Panel spaces	4 open, 2 needed	**Pass**
Circuit	60 A breaker for a 48 A continuous load (48 A x 125%)	**Pass**
Conductors	#6 AWG Cu, rated 65 A at 75°C	**Pass**
Voltage drop	1.5 V, or 0.6%, over 32 ft	**Pass**

Flagged for your review

[ ] Check the existing loads against the home. They drive the service result.

[ ] Service is at 83% of capacity. Review it before adding this load.

[ ] Hot climate: check the ambient temperature correction for conduit exposed to sun.

[ ] The town's checklist cites the 2017 NEC. Confirm the edition in force.

DEMO

Permit Application Data

Prepared for transfer to the city's permit application. This page is not the city's form.

Project address	123 Sample St, Oro Valley, AZ
Parcel number	000-00-000-000
Jurisdiction	Oro Valley, AZ
Property owner	Sample Homeowner
Permit type	Electrical, residential EV charger
Charger	Tesla Wall Connector
Contractor	Sample Electric, LLC
License number	ROC 000000

Scope of work

Install one 48 A, 240 V Level 2 EV charger (Tesla Wall Connector), hardwired, at the attached garage. Add a 60 A, 2-pole breaker to the existing 200 A main panel. Run a new dedicated 32 ft branch circuit to the charger. No service upgrade.

Inspection plan

Stage	What's inspected
Final	Panel, breaker, charger, labeling, and manufacturer instructions on site

DEMO

Site Plan

123 Sample St, Oro Valley, AZ. Parcel 000-00-000-000.



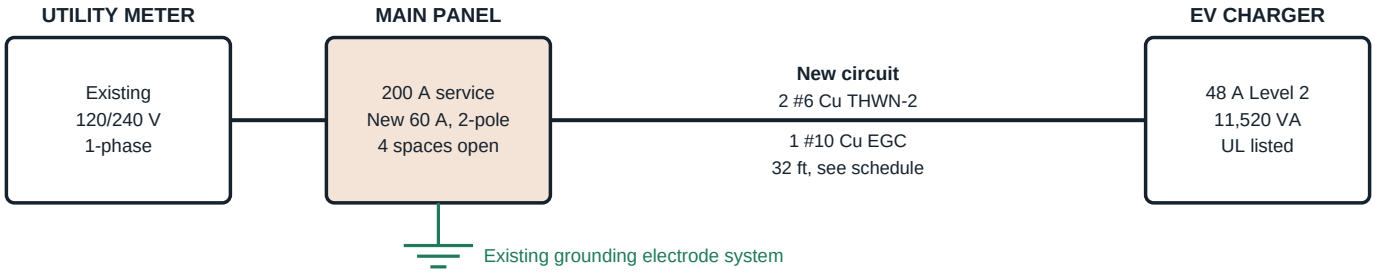
P Main panel, 200 A (existing)
EV 48 A charger (new)
New circuit route, 22 ft measured on plan
Dashed yellow: lot line

Wire run with rises: 32 ft
Charger to nearest lot line: 6 ft
Scale 1 in = 30 ft when printed at 100%
Not a survey. Field verify all dimensions.

Lot lines: Pima County GIS. Map services and data available from U.S. Geological Survey, National Geospatial Program.

One-Line Diagram

Conductors, conduit sizes and types, and overcurrent protection.



Conductor and conduit schedule

Segment	Conductors	Conduit	Protection
Above ground run	2 #6 Cu THWN-2, 1 #10 Cu EGC	3/4" EMT, 32 ft	60 A, 2-pole

DEMO

Load Calculation

Optional dwelling method, NEC 220.82, with the EV charger added at 100% of nameplate. Confirm the method the city accepts.

Load	Basis	VA
General lighting	2,100 sq ft x 3 VA	6,300
Small appliance circuits	2 x 1,500 VA	3,000
Laundry circuit	1 x 1,500 VA	1,500
Range	Nameplate	12,000
Dryer	Nameplate	5,000
Water heater	Nameplate	4,500
Dishwasher	Nameplate	1,200
Disposal	Nameplate	900
Microwave	Nameplate	1,500
Hot tub or pool	Nameplate	1,800
Subtotal		**37,700**
First 10,000 VA	100%	10,000
Remainder	27,700 VA x 40%	11,080
Air conditioning	Larger of heating or cooling, 100%	7,200
EV charger	48 A x 240 V, 100%	11,520
Total calculated load		**39,800**
Computed load	**39,800 VA / 240 V = 165.8 A**	
Existing service	200 A	
Spare capacity	34.2 A, so no upgrade is required	

DEMO

Equipment Data

Attach the manufacturer's data sheets and UL listing information behind this page.

Item	Specification	Listing
EV charger	Tesla Wall Connector, 48 A, 240 V, 11,520 VA, hardwired	UL listed, NEC Article 625
Breaker	60 A, 2-pole, matched to existing panel	Listed for the panel
Above ground conduit	3/4" EMT	Listed

DEMO

Pre-Submittal Review

Complete before sending this packet to the city.

Item	Reviewed by	Done
Existing loads match the home	Electrician	[ ]
Panel has two open spaces	Electrician	[ ]
Breaker matches the panel listing	Electrician	[ ]
Charger data sheets attached	Electrician	[ ]
HOA approval received, if required	Electrician	[ ]
Code edition confirmed with the city	Electrician	[ ]

Approval

Reviewed by	
Signature	
Date	

DEMO